

Implicit-semantic Multi-Hypothesis Dense Matching for GPS-Denied Localization in Cross-Temporal Overhead Imagery

Maciej Krupka^[0009–0000–8912–8382], Jan Wegrzynowski^[0009–0001–7901–7387], and Piotr Skrzypczynski^[0000–0002–9843–2404]

Poznan University of Technology, Institute of Robotics and Machine Intelligence,
Piotrowo 3A, 60-965 Poznan, Poland
`{firstname.lastname}@put.poznan.pl`

Abstract. Dense image matching remains fragile when images differ in scale, appearance, and acquisition time. In cross-temporal overhead imagery, seasonal change, repeated structures, and transient objects often make the strongest local correspondence geometrically wrong, while the correct match survives only as a secondary mode in the coarse matching distribution. We address this in cross-scale image-to-reference registration, where a small UAV observation must be aligned within a larger overhead reference image. We propose Sat-RoMa, which pairs a frozen DINOv3 encoder pretrained on satellite imagery—supplying a domain-matched semantic prior—with a multi-hypothesis refinement that preserves the multimodal coarse posterior instead of collapsing it into a single refined warp. For each query patch we extract several correspondence hypotheses with an adaptive-window Gaussian-mixture fit, then estimate the global homography from their union by robust geometric consensus, so that delayed commitment lets global geometry select correspondences that are locally ambiguous but mutually consistent. Counter to the usual coarse-to-fine assumption, we find that activating the network’s fine-level warp refiner increases registration error under temporal change, because it conditions on high-frequency appearance that seasonal change does not preserve. On 1500 cross-season orthophoto pairs at a 4:1 image-to-reference scale asymmetry, replacing that refiner with the multi-hypothesis stage on the same checkpoint reduces corner error more than sevenfold ($30.5 \rightarrow 4.12$ m) and raises Success within 5 m from 20.9 to 84.3%, placing Sat-RoMa ahead of every external matcher we evaluate, RoMa v2 included. The results show that posterior multimodality is a useful geometric signal for robust dense matching.

1 Introduction

Dense image matching has advanced rapidly in recent years, yet robust correspondence estimation remains difficult when the two images differ simultaneously in scale, appearance, and acquisition time. This situation is common in overhead imagery: a small current observation may need to be registered within a larger reference image acquired months or years earlier. Seasonal vegetation, crop cycles, construction, shadows, and transient objects such as parked cars can make the locally most similar region geometrically wrong, while repeated structures create several plausible correspondence modes.

This paper studies cross-temporal dense matching for overhead image registration. The application we target is UAV-to-reference-image alignment, where a small onboard image is matched against a wider georeferenced orthophoto tile. Such registration can provide an absolute correction

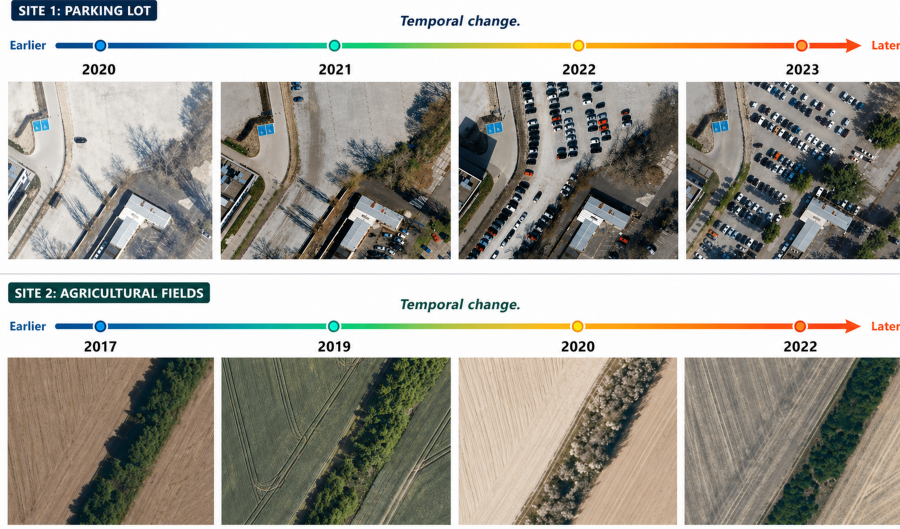


Fig. 1. Cross-temporal ambiguity in overhead imagery. **Top row:** an urban parking-lot environment with structural and moving-object changes over three years (2020–2023). **Bottom row:** an agricultural scene with strong appearance changes from seasonal vegetation cycles and crop harvesting (2017–2022).

signal for visual-inertial odometry or visual SLAM, whose local estimates drift over long trajectories. However, the central problem addressed here is not the availability of prior maps, but the visual matching ambiguity induced by temporal change. Two properties are required in this setting.

- **Temporal robustness (P1).** Query and reference images may be separated by months, years, or a full seasonal cycle. As illustrated in Fig. 1, the same place can change substantially due to vegetation growth, crop harvesting, construction, and moving objects.
- **Cross-scale robustness (P2).** The query covers only a fraction of the reference footprint; in our formulation the reference is rendered at a 4:1 linear scale ratio with respect to the query, corresponding to a 16:1 area ratio.

Our key observation is that, under such ambiguity, the correct correspondence is not necessarily absent from the matcher output. Instead, it often appears as a secondary mode in the coarse matching distribution, while the strongest mode may be locally plausible but globally inconsistent. Standard coarse-to-fine inference pipelines tend to collapse this distribution into a single refined warp before geometric verification. Once this commitment has been made, a downstream homography estimator can only reject or accept the selected correspondences; it can no longer recover alternative modes discarded by the refiner.

We therefore propose *Sat-RoMa*, a multi-hypothesis dense matching framework adapted from RoMa [7]. Rather than using the ConvNet-based fine-level warp refiner, we operate directly on the multimodal coarse matching distribution. For each query patch, an adaptive-window Gaussian-mixture fit extracts several candidate correspondence modes. The union of these candidates forms a compact multi-hypothesis correspondence set, from which a homography is estimated by RANSAC. In this way, geometric consensus performs delayed mode selection: locally ambiguous matches are retained until a globally consistent transformation can be identified.

The network architecture and training objective remain simple. We keep RoMa’s encoder-decoder but replace its DINOv2 backbone with a DINOv3 encoder pretrained on satellite imagery [15], which we use frozen; only the RoMa decoder and matching head are trained, on cross-season orthophoto pairs generated at the target 4 : 1 image-to-reference asymmetry. The frozen satellite-pretrained backbone supplies a domain-matched *implicit semantic prior*, while the novelty of the matching mechanism lies in the post-coarse inference strategy: exposing the uncertainty already present in the coarse posterior as an explicit set of geometric hypotheses, rather than suppressing it through early refinement.

Our contributions are fourfold: (i) Delayed geometric commitment from multimodal coarse matches – we treat the coarse matching distribution as a multi-hypothesis correspondence source and let homography consensus select geometrically consistent modes. (ii) Adaptive-window mode extraction – we fit local Gaussian modes with windows that expand to the estimated covariance scale, producing a compact and spatially calibrated hypothesis set. (iii) Fine-level refinement is counter-productive under temporal change – in a controlled single-location study (500 pairs), progressively activating RoMa’s FPN refiner levels monotonically *increases* corner error, with the CE median rising from 3.86 m at the coarse 1/16 stride to 7.27 m at full resolution, whereas the geometry-aware refinement on the *same* coarse distribution attains 1.61 m (median). (iv) Same-network gain and benchmark result – on 1500 cross-season orthophoto pairs at 4:1 asymmetry from four held-in locations, replacing the ConvNet fine-level refiner with the multi-hypothesis refinement on the same frozen-DINOv3 checkpoint reduces corner error more than sevenfold (30.5 → 4.12 m) and lifts Success at 5 m from 20.9 to 84.3%, placing Sat-RoMa ahead of every external matcher evaluated, RoMa v2 included.

2 Related Work

Classical local-feature pipelines detect, describe, and match sparse keypoints before geometric verification. Recent detector-free and dense matchers replace this sequence with joint image reasoning and coarse-to-fine correspondence estimation. LoFTR [16] introduced a transformer-based detector-free formulation that produces semi-dense matches without relying on repeatable keypoint detection. ASpanFormer [5] improved hierarchical attention by adapting the local search span, while Efficient LoFTR [19] revisited LoFTR for faster semi-dense matching. Dense approaches such as DKM [6] and RoMa [7] estimate dense warps and confidence maps, improving robustness under viewpoint, illumination, and scale changes. These methods provide strong correspondence priors, but their standard inference pipelines still largely refine each location towards a single correspondence or warp before global geometry is estimated.

Coarse-to-fine matching is effective because a low-resolution correlation or posterior can guide higher-resolution refinement. However, it also creates an early-commitment problem. Ambiguous structures may produce several plausible coarse modes, particularly in overhead scenes with repeated buildings, roads, field boundaries, or tree lines. RoMa is especially relevant to our work because it explicitly models coarse multimodality through anchor probabilities, yet its standard inference still uses a learned fine warp refiner that converts this uncertainty into a single dense prediction. Our method instead keeps multiple coarse modes per query patch and postpones the decision until geometric verification.

Robust estimators such as RANSAC and its later variants remain the standard mechanism for recovering geometric transformations from contaminated correspondences. Modern RANSAC-family methods, including USAC [14] and MAGSAC++ [3], improve sampling, scoring, and inlier

selection. These methods are highly effective once a candidate correspondence set has been instantiated. They do not, however, recover modes that have already been discarded by a matcher. In contrast, our use of RANSAC is geometry-facing: it operates on the union of multiple posterior modes, so robust fitting performs both outlier rejection and global mode selection. Recent work on dense match summarisation [2] is also relevant at this interface between dense correspondence and robust geometry: it reduces dense match sets to compact subsets for faster two-view estimation, whereas our aim is to retain alternative posterior modes so that geometric consensus can resolve ambiguity.

Several works have studied uncertainty in dense correspondence estimation. HD³ [22] represents hierarchical match densities, and PDC-Net+ [17] predicts constrained mixture distributions and confidence estimates for dense matching. More recent multi-hypothesis approaches such as BEAMER [18] argue that propagating several candidates across scales can avoid premature commitment. Our work is complementary: we do not introduce a new backbone for multi-hypothesis propagation. Instead, we expose the multimodal coarse posterior already produced by a dense matcher and convert it into an explicit correspondence set for homography estimation. This makes uncertainty directly usable by the geometric stage and is particularly suited to cross-temporal, cross-scale overhead registration.

Georeferenced aerial and satellite imagery has long been used as an absolute reference for outdoor localisation. Kinnari et al. [10] address GNSS-denied UAV localisation by matching onboard camera images to georeferenced orthophotos with a season-invariant Siamese CNN within a Monte-Carlo localisation framework, showing robustness to winter–summer appearance changes and perspective variation. Cross-view geo-localisation and retrieval methods [13, 20] address large appearance differences by ranking candidate tiles or predicting coarse location. Visual-inertial odometry and visual SLAM systems [4, 9, 11] provide accurate local motion estimates but accumulate drift over long trajectories [21]. These methods motivate image-to-reference registration as a downstream application. The contribution of this paper, however, is the matching mechanism itself: a delayed-commitment strategy for robust dense correspondence under temporal and scale-induced ambiguity.

3 System overview

Sat-RoMa reuses the encoder–decoder architecture of RoMa [7]: a frozen self-supervised backbone followed by a coarse-to-fine decoder that emits a dense warp field together with a per-pixel certainty map and intermediate per-patch coarse matching distributions. It deviates from the original system in four deliberate choices one architectural, two at training time, and one at inference time each tied to one of the targeted properties. Architecturally, we replace RoMa’s DINOv2 ViT-Large/14 backbone with the frozen DINOv3 ViT-L/16 encoder pretrained on SAT-493M [15], a corpus of 493M Maxar ortho-rectified RGB tiles at 0.6m ground-sample distance. Because SAT-493M is itself overhead imagery, its frozen features act as a domain-matched implicit semantic prior (P1); the backbone is used exactly as released, and only the RoMa decoder and matching head are trained.

3.1 Geometry-aware homography refinement (P1 and P2)

RoMa’s coarse decoder produces, for each query patch on a coarse grid, a categorical distribution over a discretised reference grid plus a no-match bin—a tensor in $\mathbb{R}^{(K^2+1) \times h \times w}$ which after softmax yields the per-query posterior $P(B \mid A_{x,y})$. We refer to this as the *coarse matching distribution*. In standard inference, RoMa feeds this output to a ConvNet-based fine-level warp refiner and samples a



Fig. 2. Qualitative comparison on two scenes (rows). Columns, left to right: query with its coarse patch grid (coloured by position); reference map with the ground-truth footprint (green dashed); the proposed multi-hypothesis refinement, showing the retained per-patch Gaussian modes (translucent blobs), the elected correspondences (crosses), and the recovered footprint (cyan); and the RoMa-style refiner’s single predicted keypoints with its recovered footprint (cyan). Preserving multiple hypotheses lets global consensus elect the geometrically consistent mode (cyan $\approx$ green dashed), where the single-correspondence refiner locks onto a wrong region.

single correspondence per query, followed by homography RANSAC. We instead replace this stage with a *geometry-aware homography refinement* that operates directly on the coarse distribution, bypassing the fine-level ConvNet. The proposed refinement consists of three stages.

Stage 1: adaptive Gaussian-mixture fit per posterior. For each query patch, we detect modes of $P(B | A_{x,y})$ via non-maximum suppression on a smoothed heatmap with a small dilation kernel and a low probability floor. Around each peak, we fit a 2D Gaussian $\mathcal{N}(\mu_k, \Sigma_k)$ using image moments in an *adaptively sized* window: the window expands iteratively to cover approximately $n_\sigma \approx 3$ of the fitted Gaussian. This yields tight windows for sharp peaks and wider ones for diffuse blobs, avoiding the systematic misestimation of covariance associated with fixed-window fitting.

Stage 2: multi-hypothesis correspondence set. Stage 1 produces a sparse correspondence set $\mathcal{C}$ in which each query patch contributes multiple candidate matches: $\mathcal{C} = \{(p_A^{(i)}, \mu_B^{(j)}) \mid i \in \text{patches}, j \in \text{modes}(P(B | A_i))\}$. On a 14×14 coarse grid the degenerate single-mode-per-patch case yields 196 correspondences; preserving the mixture adds the secondary modes, giving a few hundred candidates in total (e.g. ~ 256 in Fig. 2). This is far sparser than RoMa’s standard pipeline, which samples on the order of thousands of correspondences from the dense fine-level warp.

Stage 3: RANSAC consensus on the multi-hypothesis set. A standard homography RANSAC is applied to $\mathcal{C}$. Since each query patch $p_A^{(i)}$ is associated with multiple candidates $\mu_B^{(j)}$, RANSAC implicitly performs per-patch mode selection: correspondences consistent with the global geometry are retained, others rejected as outliers. The resulting homography $\hat{H}$ is reported directly, without further optimisation. The standard RoMa refinement is the degenerate case where Stage 1 produces a single Dirac per query; we report this as the *baseline* variant.

3.2 Asymmetric 4:1 image-to-map formulation (P2)

The query reproduces what a small UAV sees from low altitude: a ground footprint of only 35–70 m per edge—roughly the view from a few tens of metres up (on the order of 35–90 m with a typical onboard camera)—whereas the georeferenced prior it must localise against is a wide-area tile preloaded once for the whole mission. Localisation here is therefore intrinsically a problem of placing a small image inside a much larger map, not of aligning two views of comparable extent. Yet dense matchers in the literature are trained almost exclusively on pairs at approximately the same ground-sample distance, where both images cover similar ground and the network never has to search across scale.

We instead sample every training pair at the canonical 4:1 linear (16:1 area) image-to-map asymmetry at matched ground-sample distance: query and reference share the same resolution, but the reference spans four times the linear extent (a 224 px query against an 896 px reference tile). This makes scale-aware sub-patch localisation the actual training task—the decoder must place the small query as a sub-region of the wide reference—rather than the image-wide photometric alignment that a near-isoscale distribution rewards. As Sec. 4 shows, generic matchers tuned to near-equal scale collapse onto the wrong reference cell in exactly this regime.

3.3 Online cross-season pair synthesis

Each sample draws, in order: a location from the four-location mixture; two distinct seasons of that location; a query ground size uniformly from [35.0, 70.0] m per edge (corresponding to 5.9–11.9 cm/px at 224 px, in the low-altitude UAV regime); and a random similarity homography with rotation in $[-55^\circ, +55^\circ]$, isotropic scale in $[0.65, 1.35]$, and translation up to 30% of the tile edge. The reference is cropped at the sampled ground size from one season and rendered at 896×896 px; the query is the central 224×224 px crop of the warped tile from the other season at matched ground-sample distance.

Appearance variation between query and reference is thus drawn from *real* multi-year orthophoto acquisitions, while the geometric warp is synthetic and exactly known, providing the supervision signal for training. From 4–8 tiles per location, this process yields an effectively unbounded, geometry-labelled dataset; deterministic per-sample seeding (a stable hash of location and global sample id) ensures full reproducibility.

Fig. 2 contrasts the two pipelines on two scenes (rows); in each, left to right, we show the query with its coarse patch grid (coloured by grid position), the reference map with the ground-truth footprint (green dashed), the proposed refinement, and the RoMa-style refiner. In the proposed column the retained per-patch Gaussian modes appear as translucent blobs, the elected correspondences as crosses, and the recovered footprint in cyan; in the RoMa column we plot its single predicted key-points and recovered footprint. Top row: repeated urban structure makes several reference regions locally resemble the query, so several competing hypotheses are kept — yet geometric consensus

elects the one mutually consistent set and the cyan footprint lands on the green-dashed ground truth, while the RoMa-style refiner commits to a locally plausible but displaced region. Bottom row: the refiner’s keypoints again concentrate on a wrong location, whereas the multi-hypothesis set spreads candidates around the correct region and RANSAC recovers the right homography. In both cases, retaining the multimodal posterior is what separates a correct alignment from a confident but misplaced one.

4 Experiments and results

4.1 Experiment design

We evaluate the matching mechanism on query-to-map registration: query and reference are both orthophoto crops, the query a synthetically warped sub-tile of a different-season acquisition. Because the warp is synthetic and known, it provides exact supervision and isolates the refinement; closing the gap to real onboard-camera imagery is a deployment step we do not address here. Training data is drawn from orthophoto stacks at four 2 km×2 km locations across a Central European temperate-climate region, spanning distinct land-cover types (two dense urban centres, a peri-urban area, and forested terrain with exposed structural features), each a stack of 4–8 tiles spanning 2014–2024. Pairs are synthesised as in Sec. 3. To mirror the deployment task (matching a current observation against a historical prior over the *same* place), evaluation splits by seasonal index rather than location: the query is fixed to the newest season (2024) and candidate references are earlier acquisitions disjoint from training queries. The main evaluation therefore shares geographic extent with the training data; Sec. 4.5 additionally tests generalisation to locations and orthophoto sources that are entirely disjoint from our training data.

We compare two post-coarse refinement modes on the same Sat-RoMa checkpoint — a RoMa-style *baseline* (ConvNet fine-level refinement, then sparse-correspondence RANSAC) and the proposed *multi-hypothesis* variant (adaptive-window Gaussian-mixture fit on the coarse matching distribution, then RANSAC over the hypothesis-set union) — and against six external matchers run at their native input resolutions: LoFTR [16], LightGlue [12], ELoFTR [19], the upstream outdoor RoMa [7], RoMa v2 [8], and a classical RootSIFT [1] + RANSAC pipeline. All external matchers are evaluated using publicly released checkpoints without fine-tuning on the orthophoto distribution; for RoMa v2 in particular only the inference code is publicly available at the time of writing, so fine-tuning on the proposed cross-season pair distribution is not currently feasible without a re-implementation of its training pipeline. Each pair has a ground-truth homography H^* from rendering metadata; for every method we estimate $\hat{H}$ and report *Succ.@ τ* (fraction of pairs with mean reprojection error below τ m, $\tau \in \{5, 10, 20\}$, computed over all pairs), *Corner error* (mean ground-distance between the four query-tile corners under $\hat{H}$ vs H^* , in metres), *Scale error* ($|\sqrt{\sigma_1 \sigma_2} - 1|$ in percent, with σ_i the singular values of $J_{\text{rel}} = J_{\hat{H}}(c) J_{H^*}(c)^{-1}$ at the query centre c), and *Rotation error* (absolute rotation of J_{rel} in degrees). Corner / Scale / Rotation means are over pairs with a recovered homography. Neither Sat-RoMa variant applies post-RANSAC optimisation of $\hat{H}$.

4.2 Fine-level refinement is counterproductive under temporal change

Table 2 ablates the fine-level decoder on a single-location model (500 pairs) by progressively activating FPN refiner levels from the coarse 1/16 stride to full resolution; each configuration argmax-samples the coarse matching distribution and regresses a dense warp through the enabled levels. CE

Table 1. Comparison of matching methods on the satellite localisation benchmark (1500 samples, query:reference scale = 1:4). CE = corner error, CLE = centre localisation error, Scale = absolute scale error, S@d = success rate within d m (mean reprojection). Lower is better for CE, CLE, rotation, scale; higher is better for S@d.

Method	CE (m) ↓	CLE (m) ↓	Rot. (°) ↓	Scale (%) ↓	S@5 (%)↑	S@10 (%)↑	S@20 (%)↑
RootSIFT [1] _(1024²)	352.8	95.8	58.2	1625.9	0.1	1.0	3.8
LightGlue [12] _(1024²)	296.5	116.5	44.7	1137.4	3.4	5.2	7.5
LoFTR [16] _(832²)	386.0	85.3	31.6	380.8	0.0	0.0	0.1
ELoFTR [19] _(832²)	317.2	149.3	66.6	3695.5	0.0	0.0	0.1
RoMa (outdoor) [7] _(560²)	48.7	31.2	14.4	372.3	42.3	57.3	67.5
RoMa v2 [8] _(1280²)	13.7	9.9	7.0	11.5	77.7	84.0	86.1
Sat-RoMa (RoMa refinement) _(224², 896²)	30.5	11.5	15.1	27.0	20.9	70.0	84.0
Sat-RoMa _(224², 896²)	4.12	3.07	2.15	4.2	84.3	93.9	97.6

median increases monotonically with refiner depth: every level added to sharpen the warp instead enlarges the registration error.

This inverts the usual coarse-to-fine assumption. The fine levels regress a residual offset from local appearance correlation, which is informative only where that appearance is shared between query and reference. In the cross-season regime the high-frequency content they condition on: canopy texture, shadow geometry, crop state, transient objects is precisely the component not preserved across acquisitions, so the predicted residual is driven by temporally non-stationary evidence. Argmax-sampling the coarse matching distribution and pushing the result through the refiner therefore trades a multimodal but geometrically faithful posterior for a unimodal warp committed to locally precise, temporally unreliable peaks, and stacking levels compounds the effect.

Operating on the same coarse matching distribution with the geometry-aware homography refinement (adaptive-window Gaussian mixture, then RANSAC over the hypothesis set) instead of argmax-then-refine more than halves the best refiner configuration’s CE median. Since both consume the identical coarse input, the gap isolates the operator: the gain comes not from suppressing the fine levels but from preserving the multimodality that single-correspondence refinement, at any depth, collapses.

4.3 Encoder strength and temporal extrapolation

Table 3 crosses the two backbone variants (DINOv2, DINOv3) with two temporal generalisation splits, holding the multi-hypothesis homography refinement fixed so that the encoder is the only variable. All locations are shared with training; only the acquisitions differ. The first split pairs an unseen query image against a reference season seen during training; the second is an unseen season pair, where both acquisitions lie outside the training set. The splits thus probe increasing temporal distance from the training distribution rather than geographic novelty.

DINOv3 lowers corner error in both splits, and its margin over DINOv2 widens on the unseen season pair. The stronger encoder therefore contributes most precisely in the forward-extrapolation regime, where appearance has drifted furthest from anything seen in training. This is the opera-

tionally relevant case: deployment always matches a new acquisition against an older prior, so the gain compounds with the very temporal gap the system is built to absorb.

4.4 Results

Table 1 groups the six external matchers against the two Sat-RoMa refinement modes, which share a single checkpoint. The classical and semi-dense externals collapse at the 4:1 asymmetry: RootSIFT, LoFTR, and ELoFTR stay below 1% Success at 5 m and LightGlue reaches only 3.4%, all with corner errors of hundreds of metres, as they lock onto the wrong reference cell. The dense RoMa-family externals fare far better outdoor checkpoint RoMa reaches 42.3 / 57.3 / 67.5% Success at a 48.7 m corner error, and RoMa v2 is the strongest external at 13.7 m and 77.7 / 84.0 / 86.1%. The Sat-RoMa baseline (ConvNet fine-level refinement + RANSAC) lands between the two: it lowers corner error relative to outdoor RoMa (30.5 vs 48.7 m) but is outperformed by RoMa v2 on every metric and even trails outdoor RoMa at the tight 5 m threshold (20.9 vs 42.3%). Fine-tuning the network on the deployment distribution is therefore not, by itself, sufficient. Replacing the ConvNet refiner with the geometry-aware multi-hypothesis refinement on the *same* checkpoint takes Sat-RoMa clear of every external matcher, RoMa v2 included, reaching 4.12 m and 84.3 / 93.9 / 97.6%. Because the two Sat-RoMa rows differ only in the post-coarse operator, the 30.5 $\rightarrow$ 4.12 m gap isolates the refinement—not the backbone or the fine-tuning—as the source of the result.

On P1 — temporal robustness. Every pair is cross-seasonal by construction, so corner error and the Success columns already aggregate over temporal variation. The geometry-aware refinement lowers the Sat-RoMa baseline’s corner error by 86% (30.5 $\rightarrow$ 4.12 m) and lifts Success at 5 m from 20.9 to 84.3%. Because both rows use the same checkpoint and coarse matching distribution, the gap isolates the effect of delaying correspondence commitment to the post-coarse geometric stage: RoMa’s ConvNet fine-level warp refinement collapses its multimodality early, and any subsequent geometric verifier sees only a single committed correspondence per query. Under cross-season appearance change the dominant coarse mode is sometimes shifted by photometric ambiguity onto the wrong building or field cell while the geometrically correct mode survives as a second peak; preserving that second peak as an explicit hypothesis lets RANSAC consensus elect it.

On P2 — cross-scale robustness. Scale error drops from 27.0% to 4.2% and rotation error from 15.1° to 2.15° on the same network; the same metrics for the strongest method, RoMa v2, are 11.5% and 7.0°. The improvement is consistent with hypothesis-set RANSAC being more robust to locally ambiguous correspondences in the asymmetric pair: when a query patch can match several reference patches at different reference-side offsets (a building viewed against repeated rooflines), keeping the full mixture and letting global geometry select the consensus mode is more robust than committing to the single most likely correspondence per query. The Success rates at 10 m and 20 m (93.9 and 97.6% vs 70.0 and 84.0%) show that the gain holds across the metric range, not only in the tail.

4.5 Generalisation to an external dataset and unseen map sources

All results so far share geographic extent with training. To probe genuine out-of-distribution behaviour we evaluate, without any re-training or fine-tuning, on a fully external benchmark: the localisation dataset of Kinnari et al. [10] (SIVL). None of this data overlaps our training set—the locations are taken from a different study, and the imagery is drawn from *two distinct orthophoto sources*, neither seen during training. We keep the deployment protocol of Sec. 3: the 4:1 image-to-map asymmetry, and pairs formed by matching the newest available map against a randomly

Table 2. Effect of FPN refiner levels on corner error (single-location study, 500 pairs; CE median).

FPN refiner levels	CE median (m)↓
16	3.86
16+8	3.86
16+8+4	6.45
16+8+4+2	7.06
16+8+4+2+1	7.27
16 w/o FPN (Ours)	1.61

Table 3. Corner error (CE, metres, ↓) for two Sat-RoMa backbone variants (1500 samples each).

Evaluation split	Backbone		Gain (%)
	DINOv2	DINOv3	
Unseen query image	4.35	4.05	6.90
Unseen season pair	4.59	4.12	10.24

chosen historical acquisition, with maps spanning 2011–2024. This setting stacks three shifts at once—new geography, new map sources, and wide cross-temporal gaps—and is therefore a strict test of whether the refinement transfers beyond the training distribution rather than exploiting memorised appearance.

The original SIVL image collection could not be reproduced exactly because the licensing conditions for the Google Maps imagery used in the original study have since changed. We therefore reconstruct the benchmark using openly available orthophotos from Lantmäteriet and Maanmittauslaitos, the Swedish and Finnish national mapping authorities, respectively. Although the image sources differ from those used in [10], the geographic regions and evaluation protocol follow the same external-dataset setting. This replacement also makes the experiment reproducible using data distributed under stable public-sector licences.

Table 4 reports the result (1500 samples). Even at the tightest 5 m threshold, and on data it never saw in training, Sat-RoMa localises nearly three in four query images to within 5 m (74.4% Success), rising to 87.6 and 94.2% at 10 and 20 m; the strongest external matcher (RoMa v2) reaches only 54.5% at 5 m, and the semi-dense and classical pipelines essentially never succeed (near-zero Success, corner errors of hundreds of metres). The corner and scale errors (6.36 m, 4.98%) tell the same story. Relative to the held-in evaluation the drop is small — $S@5$ falls only from 84.3 to 74.4%, corner error from 4.12 to 6.36 m, scale error from 4.2 to 4.98% — even though the benchmark contributes entirely new geography and two unseen orthophoto sources: the system degrades gracefully rather than failing. We attribute this to the design — leaning on the frozen satellite-pretrained semantic prior and on global geometric consensus, rather than on appearance memorised from the training locations, is what lets the multi-hypothesis refinement carry over to unseen geography and sources.

5 Conclusions

The main finding is that, for cross-season image-to-map matching, preserving multimodality is more valuable than immediately refining a single dense warp: RoMa’s fine-level refiner collapses the multimodal coarse matching distribution early, and any downstream geometric verifier then inherits a single committed correspondence per query. Replacing that stage with a geometry-aware homography refinement — a Gaussian mixture fitted to each posterior, resolved into a global homography by RANSAC — reduces corner, scale, and rotation errors by roughly six- to sevenfold on the same network ($30.5 \rightarrow 4.12$ m, $27.0 \rightarrow 4.2\%$, $15.1 \rightarrow 2.15^\circ$) and lifts Success at 5 / 10 / 20 m from 20.9 /

Table 4. Generalisation to unseen areas SIVL [10]; locations and two orthophoto sources disjoint from training, newest-map-to-random-historical pairing, 4:1 scale, 1500 samples. Metrics as in Table 1.

Model	CE (m)↓	CLE (m)↓	Rot. (°)↓	Scale (%)↓	S@5 (%)↑	S@10 (%)↑	S@20 (%)↑
ELoFTR [19]	241.15	94.77	67.43	576.2	0.0	0.0	0.1
LightGlue [12]	353.61	105.52	47.97	888.3	1.5	2.1	3.3
LoFTR [16]	995.45	80.27	31.38	643.7	0.0	0.1	0.5
RoMa (outdoor) [7]	225.38	30.35	16.75	219.8	42.7	51.1	62.1
RoMa v2 [8]	46.55	45.16	22.41	310.8	54.5	60.1	64.4
Sat-RoMa (DINOv3)	6.36	4.91	3.21	4.98	74.4	87.6	94.2

70.0 / 84.0% to 84.3 / 93.9 / 97.6%, at no additional training cost. The same advantage carries to a fully external dataset with unseen geography and map sources (Sec. 4.5).

Limitations. The two design choices (multi-hypothesis correspondence set, adaptive-window mode extraction) are evaluated only jointly; isolating each is left to a follow-up. We report only the homography returned by RANSAC; a downstream Mahalanobis–Huber refinement on the Gaussian covariances is implementable from the same Stage-1 outputs but excluded pending separate validation. External matchers are run as-released, without fine-tuning on the cross-season orthophoto distribution; for RoMa v2 in particular only inference code is publicly available at the time of writing, so a fine-tuned comparison would require re-implementing its training pipeline and is left for future work. Real onboard-camera queries, strongly heterogeneous or non-nadir priors, altitude and scale regimes beyond the trained 4:1 band, incomplete or missing prior tiles, and trajectory-level use of Sat-RoMa inside a SLAM stack remain out of scope and are natural extensions.

Publicly available prior maps are an under-exploited absolute reference for outdoor robotics, but exploiting them needs a matcher that is simultaneously temporally and cross-scale robust on overhead data. Our results show that the multimodal structure of the per-query coarse matching distribution is itself a valuable source of geometric information for this task — recoverable, at no extra training cost, by geometric consensus rather than early refinement.

References

1. Arandjelović, R., Zisserman, A.: Three things everyone should know to improve object retrieval. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) (2012)
2. Astermark, J., Heyden, A., Larsson, V.: Dense match summarization for faster two-view estimation. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 1093–1102 (2025)
3. Barath, D., Matas, J., Neskova, J.: MAGSAC++: Marginalizing sample consensus. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 10197–10205 (2020). <https://doi.org/10.1109/CVPR42600.2020.00138>
4. Campos, C., Elvira, R., Rodríguez, J.J.G., Montiel, J.M.M., Tardós, J.D.: Orb-slam3: An accurate open-source library for visual, visual-inertial and multi-map slam. IEEE Transactions on Robotics **37**(6), 1874–1890 (2021)
5. Chen, H., Luo, Z., Zhou, L., Tian, Y., Zhen, M., Fang, T., McKinnon, D., Tsin, Y., Quan, L.: ASpanFormer: Detector-free image matching with adaptive span transformer. In: Proceedings of the European Conference on Computer Vision (ECCV) (2022)
6. Edstedt, J., Athanasiadis, I., Wadenbäck, M., Felsberg, M.: DKM: Dense kernelized feature matching for geometry estimation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2023)

7. Edstedt, J., Sun, Q., Bökman, G., Wadenbäck, M., Felsberg, M.: RoMa: Robust dense feature matching. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2024)
8. Edstedt, J., et al.: Roma v2: Harder, better, faster, denser feature matching. arXiv preprint arXiv:2511.15706 (2025)
9. Fontan, A., et al.: Vslam-lab: A comprehensive framework for visual slam methods and datasets. In: Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (2025)
10. Kinnari, J., Verdoja, F., Kyrki, V.: Season-invariant gnss-denied visual localization for uavs. IEEE Robotics and Automation Letters **7**(4), 10232–10239 (2022)
11. Leutenegger, S.: Okvis2: Realtime scalable visual-inertial slam with loop closure. arXiv preprint arXiv:2202.09199 (2022)
12. Lindenberger, P., Sarlin, P.E., Pollefeys, M.: Lightglue: Local feature matching at light speed. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) (2023)
13. Mall, U., Hariharan, B., Bala, K.: Change-aware sampling and contrastive learning for satellite images. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2023)
14. Raguram, R., Chum, O., Pollefeys, M., Matas, J., Frahm, J.M.: USAC: A universal framework for random sample consensus. IEEE Transactions on Pattern Analysis and Machine Intelligence **35**(8), 2022–2038 (2013). <https://doi.org/10.1109/TPAMI.2012.257>
15. Siméoni, O., Vo, H.V., Seitzer, M., et al.: DINOv3 (2025), <https://arxiv.org/abs/2508.10104>
16. Sun, J., Shen, Z., Wang, Y., Bao, H., Zhou, X.: LoFTR: Detector-free local feature matching with transformers. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 8922–8931 (2021)
17. Truong, P., Danelljan, M., Timofte, R., Van Gool, L.: PDC-Net+: Enhanced probabilistic dense correspondence network. IEEE Transactions on Pattern Analysis and Machine Intelligence (2023). <https://doi.org/10.1109/TPAMI.2023.3249225>
18. Vilain, M., Giraud, R., Berthoumieu, Y., Bourmaud, G.: Handling multiple hypotheses in coarse-to-fine dense image matching. arXiv preprint arXiv:2509.08805 (2025)
19. Wang, Y., He, X., Peng, S., Tan, D., Zhou, X.: Efficient LoFTR: semi-dense local feature matching with sparse-like speed. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 21666–21675 (2024)
20. Wu, Q., et al.: Camp: A cross-view geo-localization method using contrastive attributes mining and position-aware partitioning. IEEE Transactions on Geoscience and Remote Sensing **62**, 1–14 (2024)
21. Yao, F., et al.: Gnss-denied geolocalization of uavs using terrain-weighted constraint optimization. International Journal of Applied Earth Observation and Geoinformation **135**, 104277 (2024)
22. Yin, Z., Darrell, T., Yu, F.: Hierarchical discrete distribution decomposition for match density estimation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2019)